

# Finite-Memory CL Based on Multisensor Information Fusion Using Neural Networks for Multiple Humanoid Robots

Dong Hyeon Kim , Jung Min Pak , Peng Shi , *Fellow, IEEE*,  
and Choon Ki Ahn , *Senior Member, IEEE*

## I. INTRODUCTION

**Abstract**—Cooperative localization (CL) is a method for achieving accurate and reliable localization of multiple robots by sharing sensor measurements or position information through wireless communications among the robots and has been successfully applied to various mobile robots. However, in humanoid robots, swaying of the body caused by walking results in significant errors in LiDAR sensor measurements, degrading the accuracy of CL. To address this problem, we propose a novel CL algorithm aided by odometry information that is less affected by the swaying of humanoid robots. First, we design a finite-memory CL (FMCL) algorithm based on wireless sensor networks and LiDAR measurements that has a finite-memory structure, preventing the accumulation of errors in sensor measurements or models. Second, we process the odometry information with an artificial neural network (ANN) to obtain auxiliary position estimates. Third, the position estimates obtained by the two methods are integrated with another ANN to complete the odometry-aided FMCL (OAFMCL). Finally, we conduct CL experiments using multiple humanoid robots and demonstrate the superiority of the proposed OAFMCL over state-of-the-art localization algorithms.

**Index Terms**—Artificial neural network (ANN), cooperative localization, finite-memory structure, humanoid robot, odometry.

AS robotics technology advances, various types of robots are being developed and employed in industry. For example, wheeled robots sort packages in logistics centers, quadruped robots are used for security patrols, and humanoid robots assemble cars in automotive factories [1], [2], [3], [4]. In these modern robots, mobility and collaborative capabilities have become increasingly important [5], [6], [7]. In addition, robots need accurate localization technology to avoid collisions with other robots or humans when moving.

Cooperative localization (CL) is a method that improves the accuracy and reliability in multirobot localization scenarios by sharing information between neighboring robots. In a typical CL method, expensive, high-precision sensors are only installed in a few robots, and most of the robots are equipped with inexpensive sensors, reducing the overall system deployment cost. Each robot can send and receive information to and from its neighbors via wireless communication, allowing them to share precise measurements from the high-precision sensors or location information of nearby robots. This sharing of sensor information enables reliable localization when a particular robot's sensor measurements have large errors or even fail to acquire measurements. Although extensive research has been conducted on CL for various kinds of robots [8], [9], [10], [11], [12], [13], [14], [15], [16], few results have been reported on CL for humanoid robots.

Due to their limited applications, humanoid robots have received less attention in localization research. In [17], to compensate for camera shake caused by the bipedal gait, a technique was developed that uses the inertial measurement unit (IMU) and radio-based ranging together to process the sensor measurements through particle filtering. In [18], a lightweight humanoid robot odometric learning (LHOL) method was proposed that processed odometry information and IMU measurements with an artificial neural network (ANN). This LHOL method does not depend on vision/LiDAR sensors and can avoid measurement errors that occur in such sensors. However, localization algorithms that rely entirely on ANN can perform poorly in conditions that differ from those for which the ANN was trained.

In this study, we consider the CL of multiple humanoid robots in indoor environments in response to automobile companies

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Dong Hyeon Kim and Choon Ki Ahn are with the School of Electrical Engineering, Korea University, Seoul 13391, Republic of Korea (e-mail: cowgol99@korea.ac.kr; hironaka@korea.ac.kr).

Jung Min Pak is with the Department of Artificial Intelligence, Korea National University of Transportation, Uiwang 16106, Republic of Korea (e-mail: destin11@ut.ac.kr).

Peng Shi is with the School of Electrical and Mechanical Engineering, The University of Adelaide, Adelaide, SA 5005, Australia, and also with the Research and Innovation Centre, Obuda University, 1034 Budapest, Hungary (e-mail: peng.shi@adelaide.edu.au).

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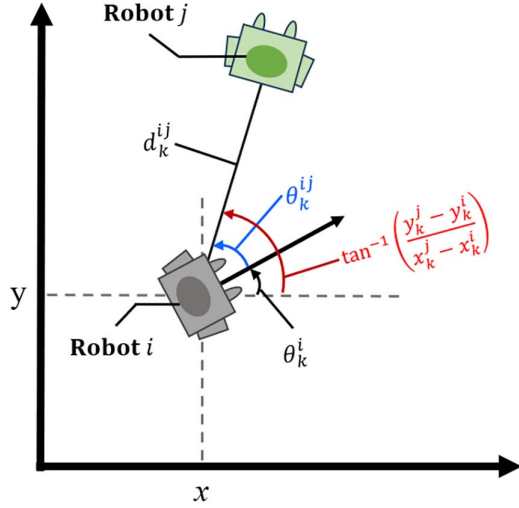


Fig. 1. 2-D geometry of relative measurements.

researching humanoid robots for use in automotive factories (e.g., Atlas for Hyundai, Fig. 1 for BMW, and Optimus for Tesla). However, measurement/model errors in humanoid robots can hinder accurate localization. For example, the bipedal gait causes the LiDAR sensors to shake, resulting in large measurement errors [19], [20], [21]. Similarly, foot slippage causes errors between the control commands and the actual movement of the robot [18], [22]. Owing to these characteristic errors, accurate CL of multiple humanoid robots is challenging.

Given these considerations, we propose a novel CL algorithm based on multisensor information fusion for humanoid robots. First, we design a new CL algorithm with a finite-memory structure to avoid the accumulation of measurement/model errors that occur in humanoid robots. The finite-memory CL (FMCL) is responsible for processing the measurements from a real-time locating system (RTLS) and LiDAR to estimate the positions of the leader and follower robots, respectively. Second, we train an ANN to process the odometry information of humanoid robots. This trained ANN takes the odometry and IMU information as input and outputs the position of the robot. Finally, another ANN is employed to integrate FMCL with odometry-based localization. The ANN responsible for the fusion is trained to take the position estimates from both localization methods as input and then outputs a unified position estimate. Each of the sensors and algorithms used for localization has drawbacks. However, our odometry-aided FMCL (OAFMCL) fuses LiDAR and odometry sensor information and combines ANN and a state estimator to overcome these drawbacks. The superior performance of the proposed algorithm is demonstrated through CL experiments using multiple humanoid robots. In the experiments, we compare our OAFMCL with state-of-the-art localization algorithms.

The novelty and contribution of our work compared with the previous CL studies can be summarized as follows. We propose a CL algorithm for humanoid robots, which has been little studied in the literature. We developed a CL algorithm based

on a robust finite-memory state estimator because humanoid robots can cause errors in ranging sensors such as LiDAR due to shaking while walking. We further improved the localization performance by fusing it with an odometry-based localization algorithm that is less affected by sensor shaking. This fusion compensates for the shortcomings of odometry-based localization, which is vulnerable to robot kidnapping. The fusion of heterogeneous sensors has been a challenging problem, but we addressed it using ANN and achieved promising performance in practical experiments.

The remainder of this article is organized as follows. In Section II, the motion/measurement models for CL are described, and a novel CL method based on the fusion of FMCL and odometry-based localization is proposed in Section III. Section IV demonstrates the performance of the proposed method through experiments. Finally, the conclusions are drawn in Section V.

## II. MOTION AND MEASUREMENT MODELS FOR COOPERATIVE LOCALIZATION

In this study, the humanoid robots are divided into two groups: a leader robot equipped with a real-time locating system (RTLS) based on wireless sensor network (WSN) technologies and follower robots equipped with low-cost LiDAR sensors. The leader robot directly estimates its absolute position by processing the distance measurements provided by the RTLS. The follower robots measure the distance and bearing of neighboring robots with LiDAR sensors, and if the coordinates of the detected neighboring robots are known, they can calculate their positions in reverse. To accomplish this, the robots are connected via a wireless communication network, allowing them to share their estimated position information with their neighbors. In this section, we describe the motion and measurement models required to perform the CL scheme.

### A. Motion Model

The motion model of a robot is an equation that describes the changes over time in the robot's position. For small humanoid robots, the error between the control input and the actual movement is mainly due to the shaking and slipping that occurs when walking. Therefore, we use a motion model that ignores control inputs to prevent control input errors from affecting localization. The constant velocity motion model [23], [24], [25] is a typical motion model without control inputs, which can be expressed as follows:

$$\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k + \mathbf{B}\mathbf{w}_k \quad (1)$$

where  $\mathbf{x}_k$  is the state vector at discrete time step  $k$ , which is defined as

$$\mathbf{x}_k = [x_k \ y_k \ \dot{x}_k \ \dot{y}_k]^T \quad (2)$$

where  $(x_k, y_k)$  and  $(\dot{x}_k, \dot{y}_k)$  are the 2-D positions and velocities, respectively. Term  $\mathbf{w}_k$  is the process noise vector, which is

assumed to be zero-mean white Gaussian noise with the covariance matrix  $\mathbf{Q}_k$ . The matrices  $\mathbf{A}$  and  $\mathbf{B}$  are defined as follows:

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & T & 0 \\ 0 & 1 & 0 & T \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} T^2/2 & 0 \\ 0 & T^2/2 \\ T & 0 \\ 0 & T \end{bmatrix} \quad (3)$$

where  $T$  is the sampling interval.

### B. Measurement Model

The measurement model is an equation that describes the relationship between measurements obtained from the sensors and state variables. Since the leader and follower robots use different types of sensors, two types of measurement models are required. First, the leader robot uses RTLS to obtain the distance measurements from three transmitters (anchors). The measurement of the distance between a leader robot and the  $a$ -th anchor is represented as follows:

$$d_k^a = \sqrt{(x_k - x^a)^2 + (y_k - y^a)^2}, \quad a \in \{1, 2, 3\} \quad (4)$$

where  $(x^a, y^a)$  is the known fixed 2-D coordinates of  $a$ -th anchor and  $(x_k, y_k)$  is the position of the leader robot. We define the measurement vector for the leader robot as  $\mathbf{z}_k = [d_k^1 d_k^2 d_k^3]^T$ . Consequently, the measurement model for the leader robot is represented as

$$\mathbf{z}_k = \mathbf{h}(\mathbf{x}_k) + \mathbf{v}_k \quad (5)$$

where  $\mathbf{h}(\cdot)$  is a nonlinear vector function, and  $\mathbf{v}_k$  is the measurement noise vector with zero-mean and covariance  $\mathbf{R}_k$ .

As shown in Fig. 1, follower robots are equipped with LiDAR sensors that measure their relative distance and bearing to neighboring robots. We denote the index number of the ego robot by  $i$  and the indexes of its neighbors by  $j$ . Consequently, the distance and bearing between **Robot**  $i$  and **Robot**  $j$  can be expressed as

$$d_k^{ij} = \sqrt{(x_k^i - x_k^j)^2 + (y_k^i - y_k^j)^2} \quad (6)$$

$$\theta_k^{ij} = \tan^{-1} \left( \frac{y_k^j - y_k^i}{x_k^j - x_k^i} \right) - \theta_k^i \quad (7)$$

where  $i$  and  $j$  indicate the index numbers of follower robots, and  $\theta_k^i$  is the heading angle of **Robot**  $i$ . This value is measured in real-time by the robot's internal sensor (i.e., IMU), and is considered a known value in the measurement model of the external sensor, LiDAR. The relative measurements in (6) can be merged using a vector form, as follows:

$$\mathbf{g}_k^{ij}(\mathbf{x}_k^i) = \begin{bmatrix} \bar{d}_k^{ij} \\ \bar{\theta}_k^{ij} \end{bmatrix} = \begin{bmatrix} \sqrt{(x_k^i - \hat{x}_k^j)^2 + (y_k^i - \hat{y}_k^j)^2} \\ \tan^{-1} \left( \frac{\hat{y}_k^j - y_k^i}{\hat{x}_k^j - x_k^i} \right) - \theta_k^i \end{bmatrix} \quad (8)$$

where the estimated 2-D location of **Robot**  $j$  is  $(\hat{x}_k^j, \hat{y}_k^j)$ , which is acquired by corresponding with **Robot**  $j$ . We establish  $\bar{\mathbf{g}}^i(\cdot) = [\mathbf{g}^{i1}(\cdot) \mathbf{g}^{i2}(\cdot) \dots \mathbf{g}^{iN_k^i}(\cdot)]^T$ , where  $N_k^i$  denotes the number of neighboring robots near to **Robot**  $i$ . Consequently, **Robot**  $i$  has the following measurement model:

$$\bar{\mathbf{z}}_k^i = \bar{\mathbf{g}}_k^i(\mathbf{x}_k^i) + \bar{\mathbf{v}}_k^i \quad (9)$$

where  $\bar{\mathbf{v}}_k^i$  is the measurement noise. We assume that the measurement noise is zero-mean white Gaussian with covariance  $\bar{\mathbf{R}}_k$ .

### C. Linearization of Measurement Model

The FMCL algorithm requires linearization of the nonlinear system models through a Taylor series expansion. The nonlinear measurement model of the leader robot (5) can be linearized using the Taylor series expansion as follows:

$$\begin{aligned} \mathbf{z}_k &\approx \mathbf{h}(\hat{\mathbf{x}}_k) + \mathbf{H}_k(\mathbf{x}_k - \hat{\mathbf{x}}_k) + \mathbf{v}_k \\ &= \mathbf{H}_k \mathbf{x}_k + [\mathbf{h}(\hat{\mathbf{x}}_k) - \mathbf{H}_k \hat{\mathbf{x}}_k] + \mathbf{v}_k \\ &= \mathbf{H}_k \mathbf{x}_k + \mathbf{e}_k + \mathbf{v}_k \end{aligned} \quad (10)$$

where  $\hat{\mathbf{x}}_k$  is the estimated state, and the Jacobian matrix  $\mathbf{H}_k$  and linearization error  $\mathbf{e}_k$  are defined as follows:

$$\mathbf{H}_k = \left. \frac{\partial \mathbf{h}_k}{\partial \mathbf{x}_k} \right|_{\hat{\mathbf{x}}_k} \quad (11)$$

$$\mathbf{e}_k = \mathbf{h}(\hat{\mathbf{x}}_k) - \mathbf{H}_k \hat{\mathbf{x}}_k. \quad (12)$$

The nonlinear measurement model (9) can also be linearized in the same way. Consequently, we obtained as

$$\bar{\mathbf{z}}_k^i = \mathbf{G}_k^i \mathbf{x}_k^i + \bar{\mathbf{e}}_k^i + \mathbf{v}_k^i \quad (13)$$

where

$$\mathbf{G}_k^i = \left. \frac{\partial \mathbf{g}_k^i}{\partial \mathbf{x}_k^i} \right|_{\hat{\mathbf{x}}_k^i} \quad (14)$$

$$\bar{\mathbf{e}}_k^i = \bar{\mathbf{g}}_k^i(\hat{\mathbf{x}}_k^i) - \mathbf{G}_k^i \hat{\mathbf{x}}_k^i. \quad (15)$$

The leader robot estimates its position using only the sensor measurements obtained through RTLS without the help of the neighboring robots. Thus, we apply the linearized measurement model (10) to the leader robot. The follower robots estimate their position by cooperating with any robots that are currently connected to them (sensing and communicating). For example, if there are currently  $N_k^i$  connected robots around **Robot**  $i$ ,  $N_k^i$  measurements can be acquired by the sensors and the estimated position information of  $N_k^i$  neighboring robots can be acquired by wireless communication. Then, model (13) can be applied.

## III. FUSION-BASED COOPERATIVE LOCALIZATION

### A. Finite-Memory Cooperative Localization

FMCL is a batch-processing algorithm that accumulates a recent finite number of measurements and processes them simultaneously. Specifically, FMCL makes an augmented measurement vector by accumulating measurement vectors in the time interval  $[m, n]$  as follows:

$$\mathbf{Z}_{m,n} = [\mathbf{z}_m^T \quad \mathbf{z}_{m+1}^T \quad \dots \quad \mathbf{z}_n^T]^T \quad (16)$$

where  $m = k - N$  and  $n = k - 1$  are the initial and final time points on the interval, respectively, and  $N$  is the memory size (window size). The state estimate of FMCL is obtained by multiplying the augmented measurement vector by a filter gain matrix as follows:

$$\hat{\mathbf{x}}_k = \mathbf{L}_{m,n} \mathbf{Z}_{m,n} \quad (17)$$

where  $\mathbf{L}_{m,n}$  is the filter gain. We now derive the filter gain of FMCL.

Term  $\mathbf{x}_k$  can be expressed as a function of  $\mathbf{x}_m$ , as follows:

$$\mathbf{x}_k = \mathbf{A}^N \mathbf{x}_m + \mathbf{F}_N \mathbf{W}_{m,n} \quad (18)$$

where

$$\mathbf{F}_N = [\mathbf{A}^{N-1}\mathbf{B} \quad \mathbf{A}^{N-2}\mathbf{B} \quad \cdots \quad \mathbf{A}\mathbf{B} \quad \mathbf{B}], \quad (19)$$

$$\mathbf{W}_{m,n} = [\mathbf{w}_m^T \quad \mathbf{w}_{m+1}^T \quad \cdots \quad \mathbf{w}_n^T]^T.$$

The augmented measurement vector can also be expressed as a function of the state at time  $m$  as follows:

$$\mathbf{Z}_{m,n} = \mathbf{T}_{m,n} \mathbf{x}_m + \mathcal{E}_{m,n} + \mathbf{U}_{m,n} \mathbf{W}_{m,n} + \mathbf{V}_{m,n} \quad (20)$$

where

$$\mathbf{T}_{m,n} = \begin{bmatrix} \mathbf{H}_m \\ \mathbf{H}_{m+1}\mathbf{A} \\ \mathbf{H}_{m+2}\mathbf{A}^2 \\ \vdots \\ \mathbf{H}_n\mathbf{A}^{N-1} \end{bmatrix},$$

$$\mathcal{E}_{m,n} = [\mathbf{e}_m^T \quad \mathbf{e}_{m+1}^T \quad \cdots \quad \mathbf{e}_n^T]^T,$$

$$\mathbf{U}_{m,n} = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} & \mathbf{0} \\ \mathbf{H}_{m+1}\mathbf{B} & \mathbf{0} & \cdots & \mathbf{0} & \mathbf{0} \\ \mathbf{H}_{m+2}\mathbf{A}\mathbf{B} & \mathbf{H}_{m+2}\mathbf{B} & \cdots & \mathbf{0} & \mathbf{0} \\ \vdots & \vdots & \cdots & \vdots & \vdots \\ \mathbf{H}_n\mathbf{A}^{N-2}\mathbf{B} & \mathbf{H}_n\mathbf{A}^{N-3}\mathbf{B} & \cdots & \mathbf{H}_n\mathbf{B} & \mathbf{0} \end{bmatrix},$$

$$\mathbf{V}_{m,n} = [\mathbf{v}_m^T \quad \mathbf{v}_{m+1}^T \quad \cdots \quad \mathbf{v}_n^T]^T. \quad (21)$$

Substituting (20) into (17) yields the following:

$$\hat{\mathbf{x}}_k = \mathbf{L}_{m,n} \mathbf{T}_{m,n} \mathbf{x}_m + \mathbf{L}_{m,n} \mathcal{E}_{m,n} + \mathbf{L}_{m,n} \mathbf{U}_{m,n} \mathbf{W}_{m,n} + \mathbf{L}_{m,n} \mathbf{V}_{m,n}. \quad (22)$$

Subtracting (18) from (22) yields the following:

$$\hat{\mathbf{x}}_k - \mathbf{x}_k = (\mathbf{L}_{m,n} \mathbf{T}_{m,n} - \mathbf{A}^N) \mathbf{x}_m + \mathbf{L}_{m,n} \mathcal{E}_{m,n} + (\mathbf{L}_{m,n} \mathbf{U}_{m,n} - \mathbf{F}_N) \mathbf{W}_{m,n} + \mathbf{L}_{m,n} \mathbf{V}_{m,n}. \quad (23)$$

Taking the expectation on both sides, we obtained as

$$\begin{aligned} \mathbb{E}[\hat{\mathbf{x}}_k - \mathbf{x}_k] &= (\mathbf{L}_{m,n} \mathbf{T}_{m,n} - \mathbf{A}^N) \mathbb{E}[\mathbf{x}_m] + \mathbf{L}_{m,n} \mathcal{E}_{m,n} \\ &\quad + (\mathbf{L}_{m,n} \mathbf{U}_{m,n} - \mathbf{F}_N) \mathbb{E}[\mathbf{W}_{m,n}] \\ &\quad + \mathbf{L}_{m,n} \mathbb{E}[\mathbf{V}_{m,n}] \\ &= (\mathbf{L}_{m,n} \mathbf{T}_{m,n} - \mathbf{A}^N) \mathbb{E}[\mathbf{x}_m] + \mathbf{L}_{m,n} \mathcal{E}_{m,n}. \end{aligned} \quad (24)$$

To obtain an unbiased estimate (i.e.,  $\mathbb{E}[\hat{\mathbf{x}}_k - \mathbf{x}_k] = 0$ ), the following two conditions should be satisfied as

$$\mathbf{L}_{m,n} \mathbf{T}_{m,n} - \mathbf{A}^N = 0 \quad (25)$$

$$\mathbf{L}_{m,n} \mathcal{E}_{m,n} = 0. \quad (26)$$

However, solutions that simultaneously satisfy (25) and (26) are generally not unique. Thus, we use (25) as a constraint and minimize  $\mathbb{E}[\hat{\mathbf{x}}_k - \mathbf{x}_k]$  to obtain the gain matrix. Assuming that condition (25) holds, we can rewrite expression (24) as

$$\mathbb{E}[\hat{\mathbf{x}}_k - \mathbf{x}_k] = \mathbf{L}_{m,n} \mathcal{E}_{m,n}. \quad (27)$$

The Frobenius norm of (27) can be represented as follows:

$$\begin{aligned} \|\mathbb{E}[\hat{\mathbf{x}}_k - \mathbf{x}_k]\|_F &= \|\mathbf{L}_{m,n} \mathcal{E}_{m,n}\|_F \\ &\leq \|\mathbf{L}_{m,n}\|_F \|\mathcal{E}_{m,n}\|_F \end{aligned} \quad (28)$$

which demonstrates that minimizing the Frobenius norm of the gain matrix also minimizes the mean estimation error. Now we set a cost function that minimizes the Frobenius norm of the gain matrix as follows:

$$J_{m,n} = \frac{1}{2} \|\mathbf{L}_{m,n}\|_F^2. \quad (29)$$

To solve the constrained optimization problem, we use the Lagrange multiplier method. The Lagrange function can be written as follows:

$$\begin{aligned} \mathcal{L}_{m,n} &= J_{m,n} - \Pi^T (\mathbf{L}_{m,n} \mathbf{T}_{m,n} - \mathbf{A}^N) \\ &= \frac{1}{2} \|\mathbf{L}_{m,n}\|_F^2 - \Pi^T (\mathbf{L}_{m,n} \mathbf{T}_{m,n} - \mathbf{A}^N) \\ &= \frac{1}{2} \text{Tr}(\mathbf{L}_{m,n}^T \mathbf{L}_{m,n}) - \Pi^T (\mathbf{L}_{m,n} \mathbf{T}_{m,n} - \mathbf{A}^N) \end{aligned} \quad (30)$$

where  $\Pi$  indicates the Lagrange multiplier. The partial derivative of  $\mathcal{L}_{m,n}$  with respect to  $\mathbf{L}_{m,n}$  should be zero as follows:

$$\frac{\partial \mathcal{L}_{m,n}}{\partial \mathbf{L}_{m,n}} = \mathbf{L}_{m,n} - \Pi^T \mathbf{T}_{m,n}^T = 0. \quad (31)$$

From (31) and (25), we obtained as

$$\Pi^T = \mathbf{A}^N (\mathbf{T}_{m,n}^T \mathbf{T}_{m,n})^{-1}. \quad (32)$$

Finally, we obtain the filter gain as

$$\mathbf{L}_{m,n} = \mathbf{A}^N (\mathbf{T}_{m,n}^T \mathbf{T}_{m,n})^{-1} \mathbf{T}_{m,n}^T. \quad (33)$$

The state estimate of FMCL can be written using the gain matrix as follows:

$$\hat{\mathbf{x}}_k = \mathbf{A}^N (\mathbf{T}_{m,n}^T \mathbf{T}_{m,n})^{-1} \mathbf{T}_{m,n}^T \mathbf{Z}_{m,n}. \quad (34)$$

The first and second elements of  $\hat{\mathbf{x}}_k$  are the estimated  $x$  and  $y$  coordinates of the robot, respectively. The FMCL of the follower robots can be obtained by replacing  $\mathbf{z}_k$  with  $\bar{\mathbf{z}}_k^i$  and  $\mathbf{H}$  with  $\mathbf{G}_k^i$ .

*Remark 1:* The memory size  $N$ , which is the length of the measurement interval, is a design parameter that affects the performance of the FMCL. As  $N$  decreases, position estimation (localization) is based on more recent information (measurements). When measurement errors are large, it is recommended to use a low value of  $N$  because the effect of the measurement error is reduced. However, if the measurement error is not severe, a large value of  $N$  is recommended because a lot of information is beneficial to suppress the measurement noise. Choosing the optimal memory size  $N$  for finite-memory estimation is possible for linear time-invariant systems, and the method has been presented in [26]. In nonlinear systems, linearization is performed at each time point, and the resulting linearized model is a time-invariant system. Therefore, a constant optimal  $N$  does not exist, and techniques are used to adaptively adjust  $N$ . However, the adaptive memory technique has the disadvantage of running multiple estimators in parallel, which is computationally expensive [27]. In this article, to



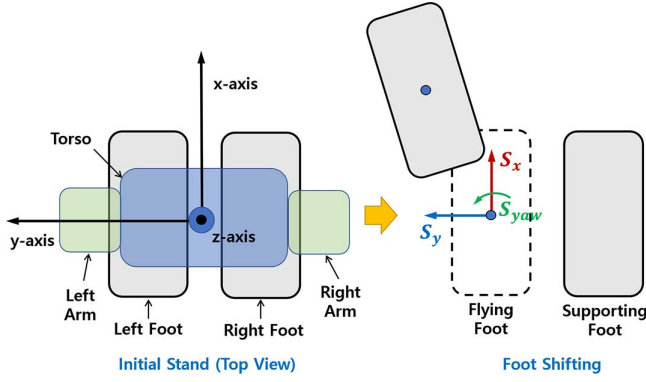


Fig. 2. Step size variables related to the 2-D movement of humanoid robot.

achieve computational efficiency, we found the best-performing memory size value through iterative simulations and used it instead of using an adaptive memory technique.

### B. Odometry-Based Localization

The odometry of a humanoid robot provides information about the robot's trajectory. This information is fed into an ANN, and the trained ANN then outputs more accurate positional information. Next, we describe the localization method with ANN using the odometry information of a humanoid robot.

A local coordinate system is set up for the gait control of a humanoid robot. The  $x$ -axis is aligned in the direction that the robot body is facing, the  $z$ -axis is perpendicular to the ground, the  $y$ -axis is perpendicular to the  $x$ - and  $z$ -axes and the origin of the coordinate system is located at the robot's hip, as shown in Fig. 2. The most important control variables that determine the robot's gait are  $S_x$ ,  $S_y$ , the step size in the  $x$ - and  $y$ -axis directions, respectively, and  $S_{yaw}$ , the rotation angle of the flying foot, which is depicted in Fig. 2.

The 3-D coordinates ( $x$ ,  $y$ , and  $z$ ) and yaw angles ( $\phi$ ) of the flying and supporting feet during one gait cycle are expressed as follows:

$$F_{\text{fly}} = \begin{cases} x = -\cos(\theta_{\text{gait}})S_x \\ y = \cos(\theta_{\text{gait}} - \pi)S_y \\ z = \exp\left(\frac{(-\theta_{\text{gait}})/Z_{\text{freq}}}{2}\right)Z_{\text{fly}} \\ \psi = \cos(\theta_{\text{gait}} - \pi)S_{\text{yaw}} \end{cases} \quad (35)$$

$$F_{\text{support}} = \begin{cases} x = -\cos(\theta_{\text{gait}} - \pi)S_x + P_{\text{PSG}}\phi_{\text{pitch}} \\ y = \sin(\theta_{\text{gait}} + \pi/T_{\text{shift}})S_y + P_{\text{RSG}}\phi_{\text{roll}} \\ z = \exp\left(\frac{(-\theta_{\text{gait}} - \pi)/Z_{\text{freq}}}{2}\right)Z_{\text{fly}} \\ \psi = \cos(\theta_{\text{gait}} - \pi)S_{\text{yaw}} \end{cases} \quad (36)$$

where  $Z_{\text{fly}}$  is the total offset of a flying leg,  $\theta_{\text{gait}}$  is the phase of the gait ( $0 < \theta_{\text{gait}} < 2\pi$ ),  $T_{\text{shift}}$  is the time shifting movement along the  $y$ -axis, and  $Z_{\text{freq}}$  determines the frequency of movement along the  $z$ -axis [18]. Additionally,  $\phi_{\text{pitch}}$  and  $\phi_{\text{roll}}$  are the outputs of IMU, which indicate the pitch and roll angles, respectively. Terms  $P_{\text{PSG}}$  and  $P_{\text{RSG}}$  are the gain values that are multiplied by the IMU's angle measurement,

which is the parameter that requires adjustment. Using the  $x$ - and  $y$ -coordinates and yaw angle of the robot's two feet, we can calculate the coordinates and heading angle of the robot at  $\theta_{\text{gait}} = 2\pi$  as follows:

$$\text{Robot}_{\text{odom}} = \begin{cases} X_{\text{odom}} = |F_{\text{fly}}(x) - F_{\text{support}}(x)|S_x \\ Y_{\text{odom}} = |F_{\text{fly}}(y) - F_{\text{support}}(y)|S_y \\ \theta_{\text{odom}} = F_{\text{fly}}(\psi) + F_{\text{support}}(\psi). \end{cases} \quad (37)$$

The information from the odometry is now fed into the ANN to train it and obtain a more accurate positioning of the robot. For training, we used the backpropagation algorithm, which calculates the error between the output of the ANN and the actual value, and then adjusts the weights as it goes backward. The ANN consists of three layers: input, hidden, and output. The input layer is a set of nodes (neurons) that receive inputs, the hidden layer computes intermediate results from the input data, and the output layer outputs the position of the robot. We used a two-layer feed-forward neural network with one hidden layer. The activation function of each node is a log-sigmoid function represented as follows:

$$\sigma(x) = \frac{1}{1 + e^{-x}}. \quad (38)$$

The final output of the ANN can be computed as

$$y = \sigma_2 \left( \sum_{i=1}^L \varpi_i \sigma_1 \left( \sum_{j=1}^S \omega_{i,j} u_j + b_1 \right) + b_2 \right) \quad (39)$$

where  $L$  is the number of nodes in the hidden layer and  $S$  is the number of inputs. Each layer is equipped with biases ( $b_1$  and  $b_2$ ), which indicate the biases of the two layers, and  $\sigma_1(\cdot)$  and  $\sigma_2(\cdot)$  are activation functions.

### C. Fusion of FMCL and Odometry-Based Localization

FMCL using exterior sensors (RTLS and LiDAR) enables high-performance localization with reasonable costs through sensor information sharing. However, the use of LiDAR in humanoid robots can be affected by the inherent motion of the robot itself, which can result in reduced accuracy. The RTLS mounted on the leader robot provides absolute positioning, which is more accurate than the LiDAR-based relative positioning used for the follower robots. However, RTLS is subject to significant variations in noise characteristics and accuracy depending on the propagation environment, including the 3-D positioning of the transmitter and receiver, obstacles, and multi-path effects. Odometry-based localization is based on the driving history according to the robot's control command, so it is less affected by the external environment or robot shaking, but it is vulnerable to problems such as robot kidnapping. Therefore, FMCL using exterior sensors and odometry-based localization can complement each other's shortcomings. We fuse FMCL and odometry-based localization using an ANN. Fusing multiple sensor information has been a challenging problem, but recent advances in ANNs have made it much easier to solve.

The configuration of the complete fusion-based localization algorithm is illustrated in Fig. 3. Since odometry-based localization and FMCL each estimate the robot's 2-D positions and

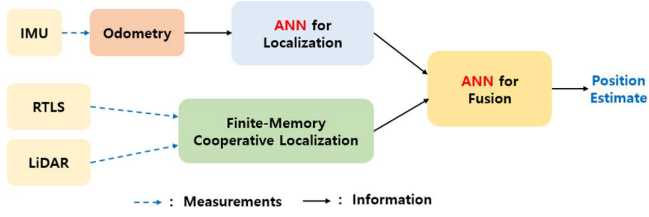


Fig. 3. Structure of fusion-based cooperative localization.

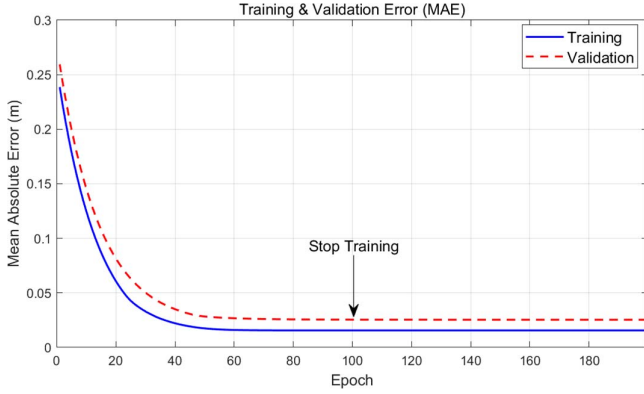


Fig. 4. Learning curves of the ANN for fusion.

velocities  $(x, y, \dot{x}, \dot{y})$ , there are a total of eight outputs from both algorithms. These values are fed into an ANN to obtain more accurate  $x$  and  $y$  coordinates of the robot. Thus, the ANN for fusion has eight inputs and two outputs. We used a two-layer feed-forward for fusion, where the number of nodes in the hidden layer was 10. The ANN for fusion uses the backpropagation algorithm to update the weights, as in the ANN for odometry-based localization.

Fig. 4 shows the learning curve of the ANN for fusion, using the mean absolute error (MAE) as the loss function, which is defined as follows:

$$\text{MAE} = \frac{1}{n_d} \sum_{i=1}^{n_d} |\hat{Y}_i - Y_i| \quad (40)$$

where  $\hat{Y}_i$  is the predicted value (i.e., ANN output) for the  $i$ -th training data sample,  $Y_i$  is the target value, and  $n_d$  is the total number of training data samples, respectively. We set the learning rate to 0.01. After 100 training iterations, we decided that no further training was needed and terminated.

#### IV. EXPERIMENTS

##### A. Experimental Setup

To verify the performance of the proposed algorithm, we conducted localization experiments using three humanoid robots. Of the three robots, one is the leader and the other two are followers. The robots used in the experiments were the BIOLOID (ROBOTIS), which were equipped with a single-board computer (Raspberry Pi4) and IMU. The leader robot was equipped with a wireless tag to receive signals from the RTLS's transmitters, and the follower robots were equipped with Cygbot's

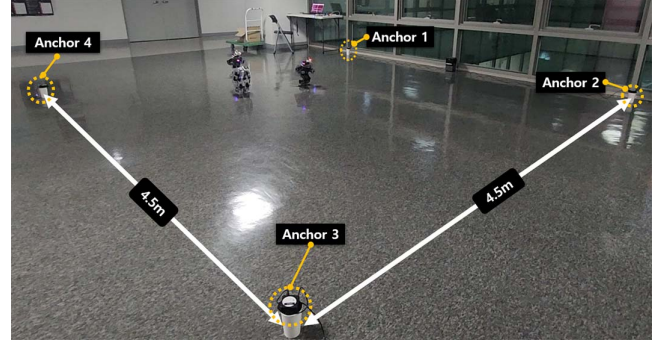


Fig. 5. Experimental environment.

low-cost LiDAR sensors. The follower robots used IMUs to measure the heading angle of their bodies and LiDAR sensors for determining their neighbors' bearings. The robots were connected by wireless networks based on WiFi to provide their estimated position information to neighboring robots. Fig. 5 displays the experimental environment, which had a smooth floor that caused the robots' feet to slip frequently, causing the robots' bodies to sway. The three humanoid robots moved around in a  $4.5 \times 4.5$  m square space. The RTLS transmitters (anchors) were made by Pozyx and were placed at every corner. If we take one of the four corners as the origin of a 2-D Cartesian coordinate system, the coordinates of all the corners are  $(0, 0)$ ,  $(0, 4.5)$ ,  $(4.5, 0)$ , and  $(4.5, 4.5)$ . Control of the robots and collection of the sensor measurements were accomplished through the robot operating system. The design parameters for the proposed algorithm were set as follows: The memory (horizon) size of the FMCL was set to  $N = 7$ , the sampling interval was set to  $T = 0.1$  s, and the process noise covariance was set to  $\mathbf{Q}_k = 0.1^2 \mathbf{I}_2$ , where  $\mathbf{I}_2$  is a  $2 \times 2$  identity matrix.

##### B. Algorithms for Comparison

We compared the proposed algorithm with three existing algorithms. The first algorithm is LHOL [18], which uses the humanoid robot's kinematics and control commands to initially estimate the robot's position by dead reckoning. This information and measurements from the IMU are then fed into an ANN to obtain a calibrated position estimate. LHOL is similar to the proposed algorithm in that it uses ANN to process odometry information to estimate position, and it is a dedicated algorithm for humanoid robot localization. Since there are few recently published humanoid robot localization algorithms other than LHOL, we considered two state-of-the-art (SOTA) algorithms for localization of wheeled mobile robots for comparative analysis. The second algorithm is the model fusion distributed Kalman filter (MFDKF) [28]. DKF is a well-known algorithm that has been widely used for multiple-robot localization, and MFDKF is a SOTA CL algorithm based on DKF. MFDKF fuses multiple measurement models to address measurement uncertainty in WSNs and improve localization accuracy. MFDKF is similar to the proposed algorithm in that it processes measurements from a WSN using a state estimator to estimate position, and it assumes a team of robots consisting

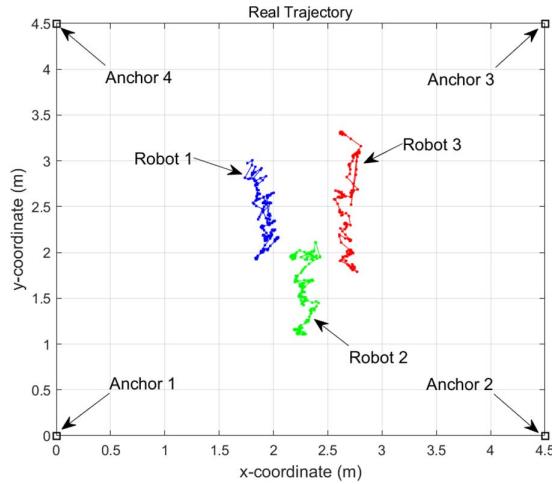


Fig. 6. Robot trajectory (ground truth) under normal condition.

of a leader and followers. The third algorithm is referred to as the nonlinear least square-based pose graph optimization (NLSPGO) [29]. NLSPGO also assumes a leader-follower structure and uses the distance and relative angle between the leader and follower robots as its measurements. NLSPGO optimizes localization results using the Levenberg-Marquardt method and the Cauchy loss function, and the result is then optimized once more by applying the sliding window PGO to obtain the final robot position estimate. NLSPGO is similar to the proposed algorithm in that it uses a leader-follower structure to perform cooperative localization and integrates multiple localization methods. We performed comparative analysis with these three existing algorithms that have similarities with the proposed algorithm. For the performance comparison, we employed the following computation for the localization error  $e_k$

$$e_k^i = \sqrt{(\hat{x}_k^i - x_k^i)^2 + (\hat{y}_k^i - y_k^i)^2} \quad (41)$$

where  $(x_k^i, y_k^i)$  and  $(\hat{x}_k^i, \hat{y}_k^i)$  are the true and estimated coordinates of the **Robot**  $i$ , respectively. In addition, we calculated the root time-averaged mean square error (RTAMSE), which is defined as

$$\text{RTAMSE}^i = \sqrt{\frac{1}{n_f} \sum_{k=1}^{n_f} \{(\hat{x}_k^i - x_k^i)^2 + (\hat{y}_k^i - y_k^i)^2\}} \quad (42)$$

where  $n_f$  indicates the final time step.

### C. Case 1: Normal Condition

We estimated the positions of three humanoid robots when they moved under normal conditions. Fig. 6 displays the ground truth of the three robots, where the four black squares in the corners represent the RTLS anchors. We used high-precision RTLS with a large number of anchors to obtain ground truth, but the small humanoid robots walked at a very slow speed, and the small distance covered by the robot relative to the magnitude of sensor noise resulted in the zigzagging trajectory in Fig. 6.

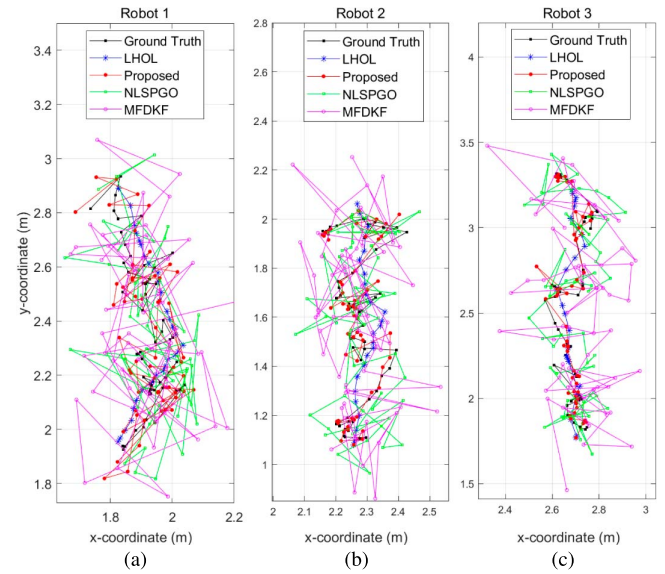


Fig. 7. Estimated trajectory under normal condition. (a) **Robot 1**. (b) **Robot 2**. (c) **Robot 3**.

**Robot 2** is the leader robot, and **Robots 1** and **3** are the follower robots.

Fig. 7 displays the estimated positions of the robots compared with the ground truth for each robot. This figure shows the degree to which the trajectories estimated by the three algorithms align with the ground truth. However, a direct comparison of the relative accuracy of the different algorithms is complicated by the presence of multiple trajectories. To facilitate a meaningful comparison of the relative performance of the localization algorithms, it is necessary to calculate the localization error and RTAMSE.

Fig. 8 displays the localization error for each robot. The localization error of the proposed algorithm was lower than that of the three existing algorithms (LHOL, NLSPGO, and MFDKF) for all three robots. The distinction between the proposed algorithm and the existing algorithms is particularly evident in the case of **Robot 2**. This is because the high-end sensor (RTLS) employed by **Robot 1**, the leader robot, yielded superior localization accuracy compared with the low-cost LiDAR utilized by the follower robots.

To analyze the localization accuracy more quantitatively, we calculated the RTAMSE, which is displayed in Table I. According to the RTAMSE values, the proposed algorithm, OAFMCL, exhibited the highest localization accuracy, followed by LHOL, NLSPGO, and MFDKF. LHOL has the best accuracy among existing algorithms because it is based on odometry information and is robust to sensor errors caused by the humanoid robot's shaking. Both NLSPGO and MFDKF do not use odometry information and rely on RTLS and IMU sensor measurements, making them more susceptible to robot shaking than LHOL. We also found that NLSPGO, which includes a localization optimization process, provides better localization on humanoid robots compared with MFDKF, which is based on fusing sensor models.



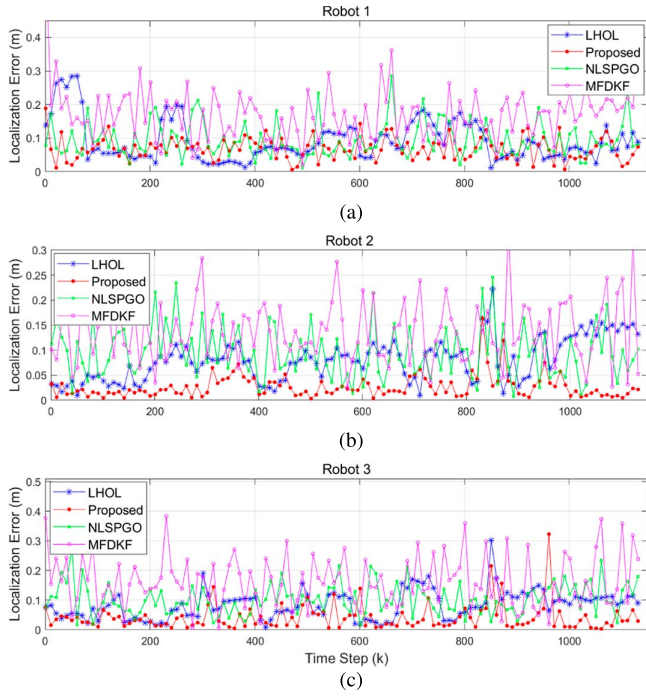


Fig. 8. Localization errors under normal condition. (a) **Robot 1**. (b) **Robot 2**. (c) **Robot 3**.

TABLE I  
RTAMSE UNDER NORMAL CONDITION (UNIT: METER)

| Algorithm                   | Robot 1 | Robot 2 | Robot 3 |
|-----------------------------|---------|---------|---------|
| MFDKF (estimation-based)    | 0.1581  | 0.1357  | 0.1642  |
| NLSPGO (optimization-based) | 0.1019  | 0.0908  | 0.1108  |
| LHOL (odometry-based)       | 0.0905  | 0.0816  | 0.0850  |
| FMCL (estimation-based)     | 0.2208  | 0.1075  | 0.1512  |
| Proposed (FMCL+odometry)    | 0.0680  | 0.0287  | 0.0456  |

Note: Blue is used to highlight the proposed algorithm. Red is used to mark the key comparative algorithms.

### D. Case 2: Robot Kidnapping

We then compared the localization accuracy of the localization algorithms in the presence of humanoid robot kidnapping. Robot kidnapping, where a small robot is picked up by a human and moved to another location, has been used in many studies as a scenario to evaluate the robustness of robot localization algorithms. In our experiment, **Robots 1, 2, and 3** were kidnapped and moved to different locations at time steps 100, 1050, and 800, respectively. Fig. 9 displays the actual trajectories (ground truth) traveled by the robots and the kidnap.

Fig. 11 presents a comparison of the position of the robot estimated by the algorithm with the ground truth. In Fig. 10(a), near the time when **Robot 1** was kidnapped ( $k = 100$ ), all of the algorithms exhibited a sharp increase in localization error. This localization error continued to increase during the time interval when the robot was lifted and moved by the human. After being placed back down on the ground, the localization error decreased rapidly for all algorithms except LHOL. LHOL relies on odometry information calculated from the robot's control commands, which makes it vulnerable to deviations from the control commands. On the other hand, NLSPGO and MFDKF,

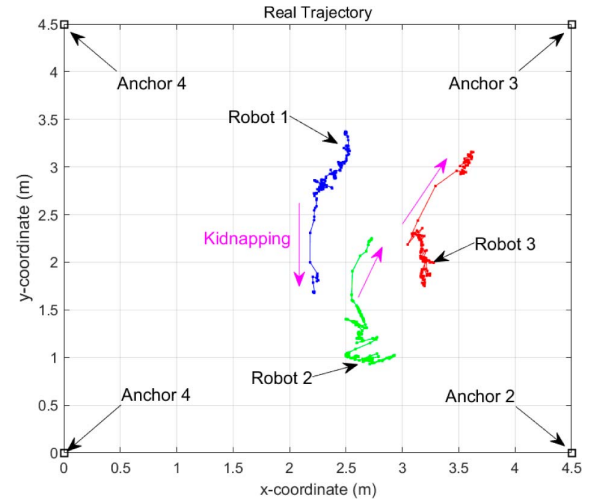


Fig. 9. Robot trajectory in case of robot kidnapping.

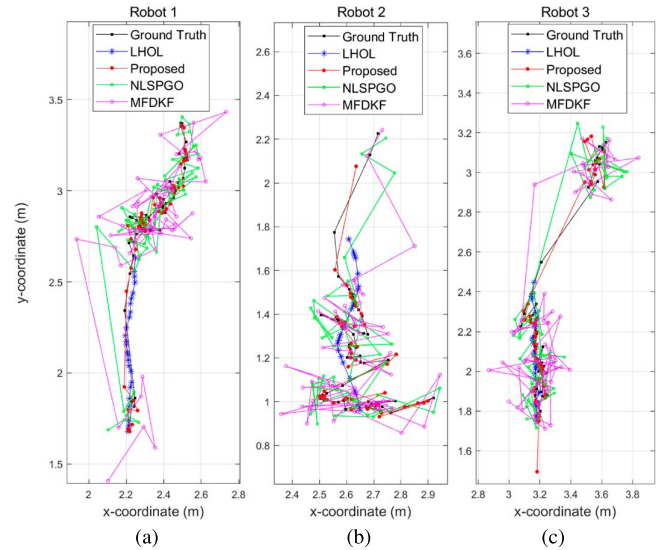


Fig. 10. Estimated trajectory in case of robot kidnapping. (a) **Robot 1**. (b) **Robot 2**. (c) **Robot 3**.

which rely entirely on sensor information, are robust to robot kidnapping. The proposed algorithm OAFMCL fuses odometry information with sensor measurements and is therefore robust to both robot kidnapping and sensor errors due to robot shaking. Therefore, it showed better localization accuracy than the three existing algorithms under robot kidnapping as well as under normal conditions. For **Robots 2 and 3**, it was also evident that the localization error increased sharply near the time of the kidnapping (1050 and 800, respectively). After the robot kidnapping, the other three algorithms, except LHOL, showed a rapid decrease in localization error.

Table II provides a comparison of the RTAMSE values of the localization algorithms in the kidnapping scenario. LHOL, being vulnerable to robot kidnapping, produced the highest RTAMSE values, while the proposed algorithm produced the lowest RTAMSE values. NLSPGO with optimization produces lower RTAMSE values than MFDKF even in the robot kidnapping scenario.



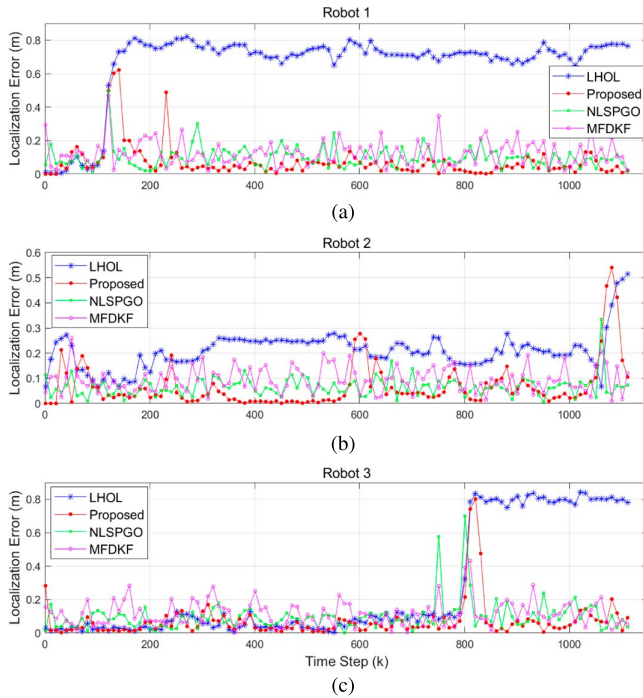


Fig. 11. Localization errors in case of robot kidnapping. (a) Robot 1. (b) Robot 2. (c) Robot 3.

TABLE II  
RTAMSE IN CASE OF ROBOT KIDNAPPING (UNIT: METER)

| Algorithm                   | Robot 1 | Robot 2 | Robot 3 |
|-----------------------------|---------|---------|---------|
| MFDKF (estimation-based)    | 0.1250  | 0.0900  | 0.1158  |
| NLSPGO (optimization-based) | 0.1026  | 0.0672  | 0.0946  |
| LHOL (odometry-based)       | 0.6605  | 0.2108  | 0.2661  |
| FMCL (estimation-based)     | 0.2239  | 0.1113  | 0.1876  |
| Proposed (FMCL+odometry)    | 0.0697  | 0.0661  | 0.0703  |

Note: Blue is used to highlight the proposed algorithm. Red is used to mark the key comparative algorithms.

### E. Comparison of Computational Complexity

Table III shows the computation time for a single estimate, computational complexity, and space complexity (memory consumption) of the localization algorithms. The computation times were measured on a computer with an AMD Ryzen 5 5600X (6 Core, 3.70 GHz). In the expression for computational and space complexity,  $p$  is the dimension of the state vector,  $N$  is the memory size for finite-memory estimation,  $W$  is the total number of weights in the neural network, and  $h$  is the number of iterations of the optimization. A comparison of the time taken by the localization algorithms for a single estimate reveals that MFDKF required the least time, while NLSPGO required the most. The proposed algorithm has a longer computation time than both MFDKF and LHOL, but it is much shorter than NLSPGO. Since the proposed algorithm fuses two kinds of algorithms, it takes more computation time than either MFDKF or LHOL, which are single algorithms. However, considering that MFDKF and LHOL have the shortcomings of large localization errors in normal and kidnapping conditions, respectively, the proposed algorithm achieves good accuracy in both conditions at the

TABLE III  
COMPARISON OF COMPUTATIONAL COMPLEXITY

| Algorithm | Computation Time ( $\mu s$ ) | Computational Complexity | Space Complexity   |
|-----------|------------------------------|--------------------------|--------------------|
| MFDKF     | 26.05                        | $O(p^2)$                 | $O(p^2)$           |
| LHOL      | 73.07                        | $O(W)$                   | $O(W)$             |
| Proposed  | 155.61                       | $O(N \cdot p^2 + W)$     | $O(N \cdot p + W)$ |
| NLSPGO    | 668.31                       | $O(h \cdot l^2)$         | $O(l)$             |

Note: Blue is used to highlight the proposed algorithm

expense of computation, which is reasonable. The proposed algorithm also showed very low computation compared with NLSPGO, the state-of-the-art algorithm that performed well in both conditions. In addition, the computation time required for a single estimate of the proposed algorithm is 155  $\mu s$ , which is only about 6500th of a second compared with the sampling interval of 0.1 s, allowing for real-time processing.

### V. CONCLUSION

In this article, the OAFMCL algorithm was proposed for accurate and reliable localization of multiple humanoid robots. Algorithms that use external sensors such as WSNs or LiDAR for humanoid robot localization are vulnerable to shaking due to robot walking or changes in the environment. The localization algorithm based on robot odometry information developed to solve this problem is vulnerable to robot kidnapping. The proposed algorithm, OAFMCL, achieves a synergistic effect by processing and fusing information from two types of complementary sensors through FMCL and ANN. Through experiments, OAFMCL has demonstrated superior accuracy and reliability compared with the state-of-the-art localization algorithms, and it is computationally efficient and can be implemented in real-time.

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**Dong Hyeon Kim** received the B.S. degree in robotics from Kwangwoon University, Seoul, South Korea, in 2022.

He is currently a Graduate Researcher with the School of Electrical Engineering, Korea University, Seoul. His research interests include multirobot systems, reliable localization, distributed control and estimation, and finite impulse response filtering.



**Jung Min Pak** received the B.S., M.S., and Ph.D. degrees in electrical engineering from Korea University, Seoul, South Korea, in 2006, 2008, and 2015, respectively.

He was an Assistant Professor with Wonkwang University, Iksan, South Korea, from 2016 to 2020 and an Associate Professor from 2020 to 2023. He is currently an Associate Professor with the Department of Artificial Intelligence, Korea National University of Transportation, Uiwang, South Korea. His

research interests include state estimation, target tracking, and localization.



**Peng Shi** (Fellow, IEEE) received the Ph.D. degree in electrical engineering from the University of Newcastle, Newcastle, NSW, Australia, in 1994, the Ph.D. degree in mathematics from the University of South Australia, Adelaide, SA, Australia, in 1998, the D.Sc. degree from the University of Glamorgan, Pontypridd, U.K., in 2006, and the D.Eng. degree from The University of Adelaide, Adelaide, SA, Australia, in 2015.

He is currently a Professor with The University of Adelaide. His research interests include systems and control theory and applications to autonomous and robotic systems, network systems, and cyber-physical systems.

Dr. Shi is the Editor-in-Chief of *Transactions on Cybernetics*, the Senior Editor of *IEEE ACCESS*, the Distinguished Lecturer of IEEE SMC Society, and the President of the *International Academy for Systems and Cybernetic Science*. He is a fellow of IET, IETI, IEAust, CAA and a member of the Academy of Europe.



**Choon Ki Ahn** (Senior Member, IEEE) received the Ph.D. degree in control engineering from the School of Electrical Engineering and Computer Science, Seoul National University, Seoul, Korea, in 2006.

Dr. Ahn is currently a Distinguished Chair Professor with the School of Electrical Engineering, Korea University, Seoul. In 2017, he received the Presidential Young Scientist Award from the President of South Korea. In 2020, 2021, and 2023, he received the Best (Outstanding) Associate Editor Award for IEEE Transactions on Neural Networks and Learning Systems; IEEE Transactions on Systems, Man, and Cybernetics: Systems; and IEEE Transactions on Circuits and Systems I: Regular Papers, respectively. From 2019 to 2025, he received the Korea University Research Excellence Award. In 2023, he was inducted into the Korea University Hall of Fame as the world's top researcher. He has been a Senior Editor of IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON SYSTEMS, MAN, AND CYBERNETICS: SYSTEMS, and IEEE SYSTEMS JOURNAL. He has also been an Associate Editor of IEEE TRANSACTIONS ON FUZZY SYSTEMS, IEEE TRANSACTIONS ON AUTOMATION SCIENCE AND ENGINEERING, IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS, IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I: REGULAR PAPERS, and *IEEE Systems, Man, and Cybernetics Magazine*; and other flagship journals. He is the recipient of the 2019–2024 Highly Cited Researcher Award in Engineering by Clarivate Analytics (formerly Thomson Reuters).